

# Toyota Case



C A T A R I N A  
S A R M E N T O



911 emergency, what are you reporting?

Our accelerators stuck!

I'm sorry?

Our accelerators stuck!

We're going 120, Mission Gorge!

We're ...we're in trouble

...There's no brake!







# Malcom Gladwell

“IF YOUR CAR IS SUDDENLY AND MYSTERIOUSLY  
ACCELERATING, ALL YOU HAVE TO DO IS STEP ON  
THE BRAKES BECAUSE BRAKES BET ENGINES.”



# Who's to blame?



- × 11,000 global variables;
  - × “bolo rei” code;
  - × 67 functions were rated untestable;
  - × absence of any bug-tracking system;
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- × MISRA-C coding 93 required and 34 advisory rules → Toyota nailed six of them;
  - × Total: 80,000 rule violations.

# What about Tesla?

12 / 2019 – ELECTRONIC BRAKES BEGIN



# Responsibility





